

Blink of LED

```
void setup() {  
  // initialize digital pin LED_BUILTIN as an output.  
  pinMode(LED_BUILTIN, OUTPUT);  
}  
  
// the loop function runs over and over again forever  
void loop() {  
  digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)  
  delay(1000);                    // wait for a second  
  digitalWrite(LED_BUILTIN, LOW);  // turn the LED off by making the voltage LOW  
  delay(1000);                    // wait for a second
```

}LED Fading

```
int led = 9;      // the PWM pin the LED is attached to  
int brightness = 0; // how bright the LED is  
int fadeAmount = 5; // how many points to fade the LED by  
  
// the setup routine runs once when you press reset:  
void setup() {  
  // declare pin 9 to be an output:  
  pinMode(led, OUTPUT);  
}  
  
// the loop routine runs over and over again forever:  
void loop() {  
  // set the brightness of pin 9:  
  analogWrite(led, brightness);
```

```

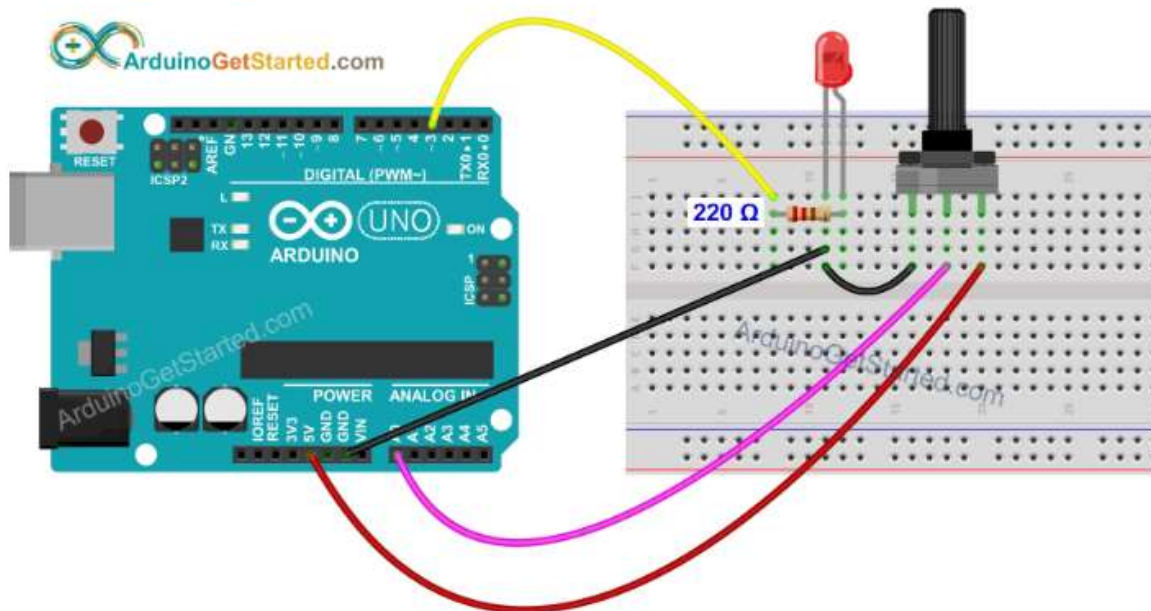
// change the brightness for next time through the loop:
brightness = brightness + fadeAmount;

// reverse the direction of the fading at the ends of the fade:
if (brightness <= 0 || brightness >= 255) {
  fadeAmount = -fadeAmount;
}
// wait for 30 milliseconds to see the dimming effect
delay(30);
}

```

LED fading using Potentiometer

Wiring Diagram



```
int LED_PIN = 3; // the PWM pin the LED is attached to
```

```
// the setup routine runs once when you press reset:
```

```
void setup() {
```

```

// initialize serial communication at 9600 bits per second:
Serial.begin(9600);

// declare LED pin to be an output:
pinMode(LED_PIN, OUTPUT);
}

// the loop routine runs over and over again forever:
void loop() {
  // reads the input on analog pin A0 (value between 0 and 1023)
  int analogValue = analogRead(A0);

  // scales it to brightness (value between 0 and 255)
  int brightness = map(analogValue, 0, 1023, 0, 255);

  // sets the brightness LED that connects to pin 3
  analogWrite(LED_PIN, brightness);

  // print out the value
  Serial.print("Analog: ");
  Serial.print(analogValue);
  Serial.print(", Brightness: ");
  Serial.println(brightness);
  delay(100);
}

```

Ultrasonic sensor

```

const int trigPin = 9;
const int echoPin = 10;

```

```
// defines variables

long duration;

int distance;


void setup()

{

pinMode(trigPin, OUTPUT); // Sets the trigPin as an Output
pinMode(echoPin, INPUT); // Sets the echoPin as an Input
Serial.begin(9600); // Starts the serial communication
}


// the loop function runs over and over again forever
void loop()

{

    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    // Sets the trigPin on HIGH state for 10 micro seconds
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);
    // Reads the echoPin, returns the sound wave travel time in microseconds
    duration = pulseIn(echoPin, HIGH);
    // Calculating the distance
    distance= duration*0.034/2;
    // Prints the distance on the Serial Monitor

    Serial.println(distance);

    delay(100);

}
```

